

LAB 16 RESULTS

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OVERVIEW

Objective

- Understand the differences and advantages of single-ended vs differential measurements
- Use an Arduino alongside a 16-bit ADS1115 board and TMP36 temperature sensor

Measurements

- Measuring the difference between single-ended and differential measurement methods by injecting a small amount of noise in the ground rail. This is done by using a 5Hz and $20V_{pk-pk}$ Sin wave from a function generator.

FINDINGS

An Arduino, ADS1115, and TMP36 temperature sensor were wired to create a circuit to plot a single-ended and differential measurement. The ADS1115 is used as it is 16 bits, a higher resolution than the Arduino, to convert the TMP36's analog signal to digital.

Single-Ended	Differential
Affected by ground noise	Not affected by injected ground noise
Utilizes one ground trace	Requires extra traces
Saves PCB space but can introduce noise	Takes up PCB space but is good for isolating signals

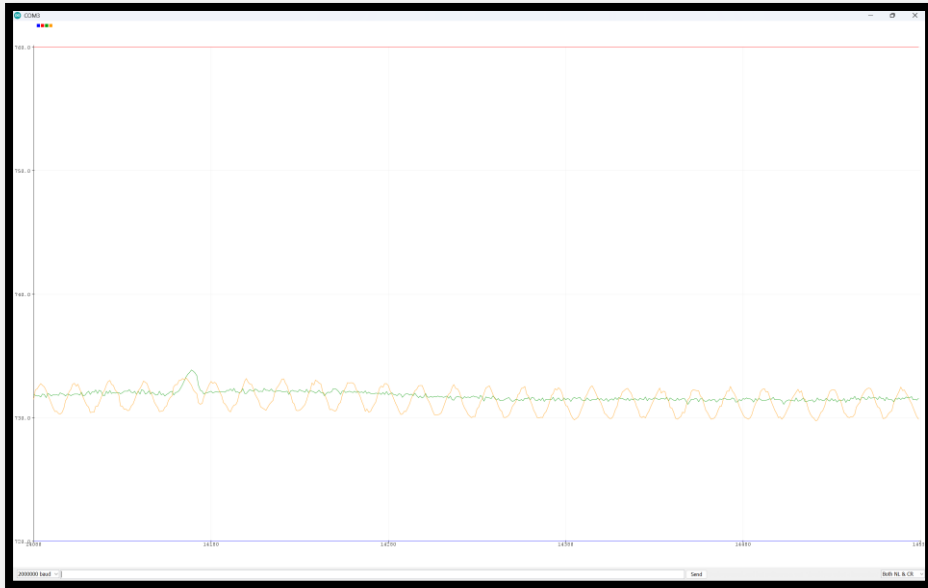
- Both have their advantages and disadvantages
- Differential traces are usually used in important applications such as video or audio outputs where minimizing distortion is crucial
- If a measurement is being taken over a quiet high and low, single-ended will be just fine.

I2C PULL UP RESISTORS

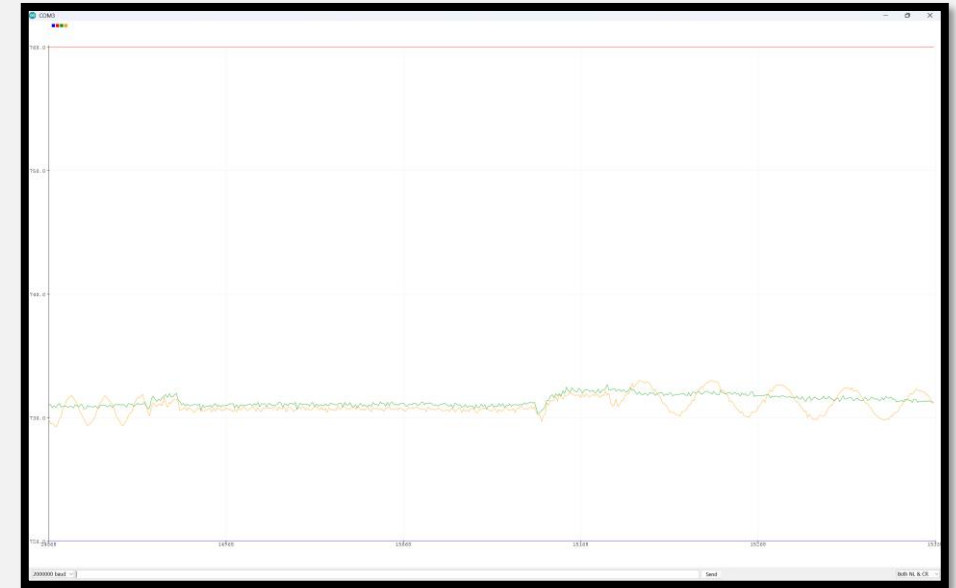
Pull up resistors are useful as they stop pins from “floating” in a circuit. They always provide a default high state if the pin is not actively being used. This prevents noise from leaking through and keeps the pin at a relatively constant voltage.

- A pull up resistor's value cannot be too low or current might leak through
- A pull up resistor's value cannot be too high or the circuit's speed will diminish
- I will be using a $1k\Omega$ resistor on my board 3 Arduino design. A good in-between value

OSCILLOSCOPE/CIRCUIT IMAGES



Singled-Ended (Orange) and Differential (Green) Scope 0
5 Hz 20V_{pk-pk} Sin Wave



Singled-Ended (Orange) and Differential (Green) Scope 1
Switching to 2.5 Hz 20V_{pk-pk} Sin Wave